

$\Sigma = \{m, ' ', (,)\}$ $V = \{S, L, F, N, E\}$

$S \rightarrow L$

$L \rightarrow (F)$

$F \rightarrow \epsilon$

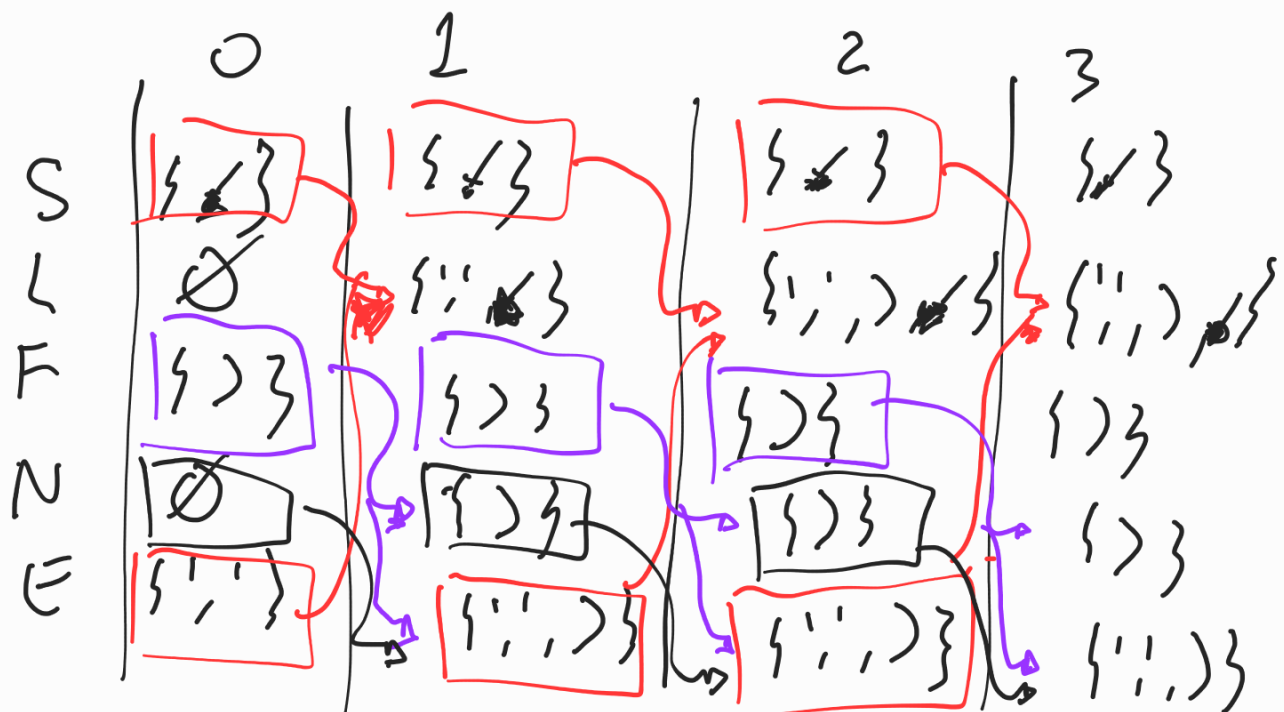
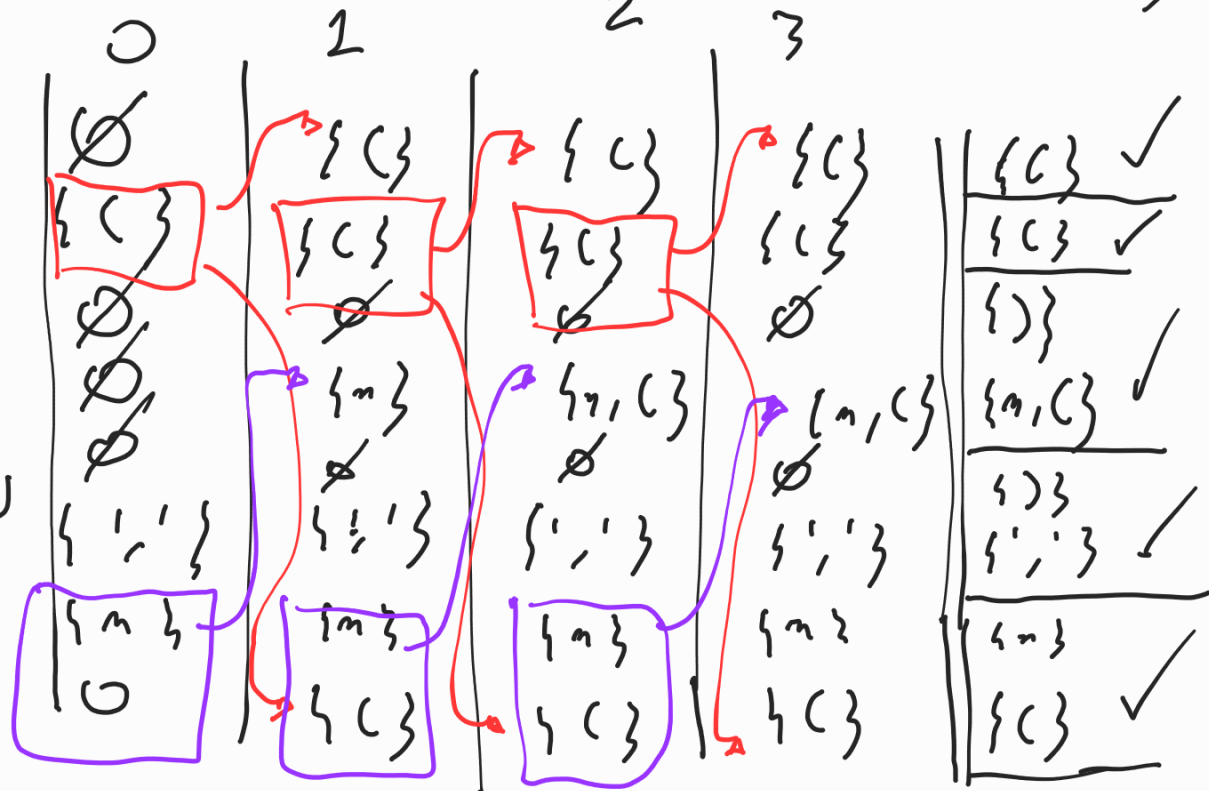
$F \rightarrow \epsilon N$

$N \rightarrow \epsilon$

$N \rightarrow ' ' \epsilon N$

$\epsilon \rightarrow m$

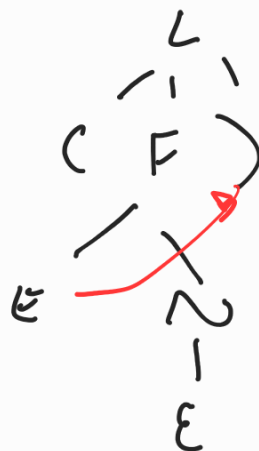
$\epsilon \rightarrow L$



$\epsilon \rightarrow L L (2)$

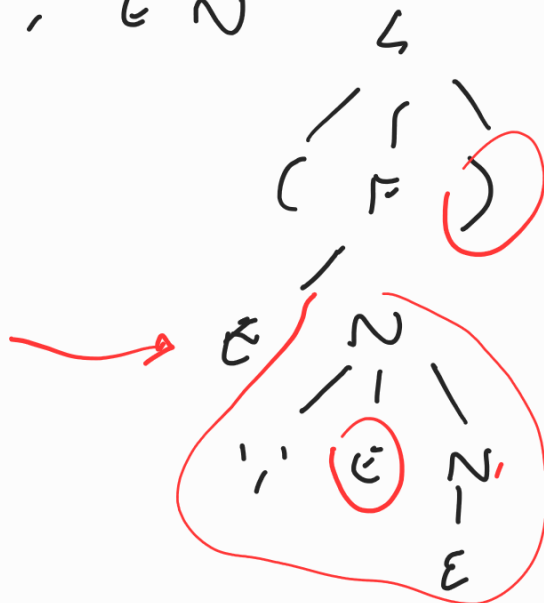
$F \rightarrow \epsilon N$

(F)



$(\epsilon F) \rightarrow (\epsilon)$

$N \rightarrow ' , ' \epsilon N$



$(\epsilon ' , ' \epsilon) \epsilon$

$$\Sigma = \{i, v, n, d\} \quad V = \{S, R, L, X\}$$

	0	1	2	GV1
$S \rightarrow R i S$	$\emptyset$	$\{i\}$	$\{i\}$	$\{i\}$ ✓
$S \rightarrow \epsilon$	$\emptyset$	$\emptyset$	$\emptyset$	$\{ \}$
$R \rightarrow i L v$	$\{i\}$	$\{i\}$	$\{i\}$	$\{i\}$ ✓
$L \rightarrow d X R$	$\{d\}$	$\{d\}$	$\{d\}$	$\{d\}$
$L \rightarrow n$	$\{n\}$	$\{n\}$	$\{n\}$	$\{n\}$ ✓
$X \rightarrow n X$	$\{n\}$	$\{n\}$	$\{n\}$	$\{n\}$
$X \rightarrow \epsilon$	$\emptyset$	$\emptyset$	$\emptyset$	$\{i\}$ ✓

	0	1	2
S	$\{ \}$	$\{ \}$	$\{ \}$
R	$\{i\}$	$\{i, v\}$	$\{i, v\}$
L	$\{v\}$	$\{v\}$	$\{v\}$
X	$\{i\}$	$\{i\}$	$\{i\}$

$$\in L(1)$$

[illegible]

$$\Sigma = \{n, +, *, (,)\}$$

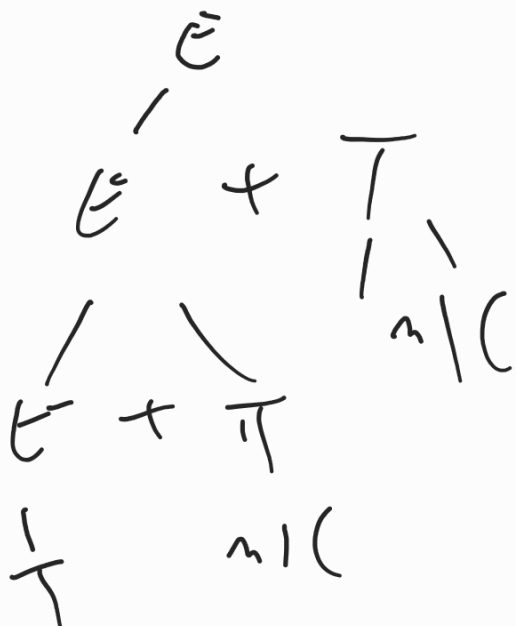
$$V \approx \{s, e, T, F\}$$

$S \rightarrow E$
 $E \rightarrow E + T$
 $E \rightarrow T$
 $T \rightarrow T * F$
 $T \rightarrow F$
 $F \rightarrow m$
 $F \rightarrow (E)$

0 1 2 3 4 = Gini

State 0: $\emptyset$
 State 1: $\emptyset$
 State 2: $\emptyset$
 State 3: $\{m, \{ \}$
 State 4: $\{m, \{ \}$

Transitions:
 - Red arrows (E): 0 → 1, 1 → 2, 2 → 3, 3 → 4
 - Purple arrows (T): 0 → 1, 1 → 2, 2 → 3, 3 → 4



EXPRESSION LOGIC

$$V = \{S, EL, TL, FL, PC, E, T, F\}$$

$$\Sigma = \{AND, OR, NOT, (,), +, *, ^, \div\}$$

$$S \rightarrow EL$$

$$EL \rightarrow EL \text{ OR } TL$$

$$EL \rightarrow TL$$

$$TL \rightarrow TL \text{ AND } FL$$

$$TL \rightarrow FL$$

$$FL \rightarrow NOT(EL)$$

$$FL \rightarrow (EL)$$

$$FL \rightarrow PC$$

$$PC \rightarrow E \text{ or } E$$

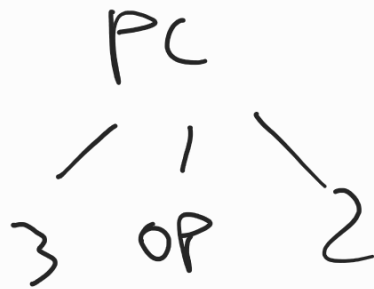
$lex \in \mathcal{L}$

$op : (">" | ">=" | "<" | "<=" | "==" | "!=")$



$3 > 2$

$3 == 2$



$opgt : ">"$
 $opgte : ">="$
 $oplt : "<"$
 $oplte : "<="$
 $opeq : "=="$
 $opneq : "!="$

$PC \rightarrow E \ opgt \ E$
 $PC \rightarrow E \ opgte \ E$
 $PC \rightarrow E \ oplt \ E$
 $PC \rightarrow E \ oplte \ E$
 $PC \rightarrow E \ opeq \ E$
 $PC \rightarrow E \ opneq \ E$